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- ⦿ Connectivity GPIB Instrument via USB

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- ✦ USB 2.0 compatible
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Q. What is a GPS system? How does it work?

Azariah

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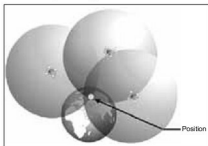
A. The global positioning system (GPS), also known as NAVSTAR, is a satellite-based navigation system that uses radio signals and provides reliable positioning, navigation and timing services to worldwide users irrespective of weather conditions, day and night, anywhere on the earth.

Using the GPS, following two values can be determined anywhere on the Earth:

1. One's exact location (longitude, latitude and height coordinates) accurate to within a range of 20 metres to about 1 mm

2. The precise time (Universal Time Coordinated (UTC)) accurate within a range of 60 ns to about 5 ns

The coordinates and time values are determined by 28 satellites orbiting the Earth. Velocity and direction of travel (course) can be derived from



Determination of location on the ground

these coordinates as well as the time. The GPS receiver can calculate other information, such as bearing, track, trip distance, distance to destination, and sunrise and sunset time.

How does it work? GPS technology consists of three sections: space, control and user.

At least 28 GPS satellites orbit the earth twice a day in specific pattern. These satellites travel at approximately 11,265 kmph (7000 miles per hour) about 19,312 km (12,000 miles) above the earth's surface. They are spaced such that a GPS receiver anywhere in the world can receive signals from at least four of them. Each GPS satellite

constantly sends coded radio signals to the earth.

The control section is responsible for constantly monitoring the satellites' health, signal integrity and orbital information from the ground. It uploads the information to satellites to provide maximum user accuracy.

The GPS user segment consists of GPS receiver. The GPS receiver collects and processes the signals from the GPS satellites that are in view and then uses that information to determine and display on a screen the current location on the ground, velocity and time. It does not send any information back to the satellites.

All of the GPS satellites and the receiver have an internal clock. Each satellite transmits a coded signal. This code is generated as a function of time. The receiver is also able to generate the same code. The receiver matches the incoming code to the internally generated code and computes the time taken by the signal to reach the receiver and the traveling velocity of the signal. Next, it can calculate the distance from the satellite.

If you know how far you are from one satellite, you know that you are somewhere along an imaginary sphere around that satellite. If you know how far you are from two satellites, you are somewhere along the intersection of these two spheres, which is a circle. If you add another satellite, you are where this third sphere intercepts the circle created by the intersection of the other two spheres (see the figure). The sphere will most likely intercept the previous circle at two points. One of these points is where you are and the other is not a reasonable solution—somewhere in outer space. Thus by knowing where you are relative to these three satellites, the receiver with a perfect clock can know where it is.

Answers compiled by Col (Retd) N.C. Pande

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